

File Edit Selection View Go ... Streamlit

EXPLORER

STREAMLIT

0.Image analysis

final_app.py

np_image.py

np1_image.py

code1.py

firstapp.py

myimage.py

myimage.py > ...

```
1 import streamlit as st
2 import numpy as np
3 from PIL import Image
4 import requests
5 from io import BytesIO
6 import matplotlib.pyplot as plt
7
8 # Set Streamlit page config
9 st.set_page_config(page_title="Priyanka Sulaganti Image Processor", layout="wide")
10
11 # Title
12 st.title("Priyanka Sulaganti Image - Multi-Color Channel Visualizer")
13
14 # Load image from local path
15 @st.cache_data
16 def load_image():
17
18     path=r"C:\Users\DELL\Downloads\Priya.jpg"
19
20     return Image.open(path).convert("RGB")
21
22 # Load and display image
23 priyanka = load_image()
24 st.image(priyanka, caption="Original Priyanka Image", use_container_width=True)
25
26 # Convert to NumPy array
27 priyanka_np = np.array(priyanka)
28 R, G, B = priyanka_np[:, :, 0], priyanka_np[:, :, 1], priyanka_np[:, :, 2]
29
30 # Create channel images
```

> OUTLINE

> TIMELINE

02:03 PM

Visual Studio Code interface showing a Streamlit application named `myimage.py`.

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```
30 # Create channel images
31 red_img = np.zeros_like(priyanka_np)
32 green_img = np.zeros_like(priyanka_np)
33 blue_img = np.zeros_like(priyanka_np)
34
35 red_img[:, :, 0] = R
36 green_img[:, :, 1] = G
37 blue_img[:, :, 2] = B
38
39 # Display RGB channels
40 st.subheader("RGB Channel Visualization")
41 col1, col2, col3 = st.columns(3)
42
43 with col1:
44     st.image(red_img, caption="Red Channel", use_container_width=True)
45
46 with col2:
47     st.image(green_img, caption="Green Channel", use_container_width=True)
48
49 with col3:
50     st.image(blue_img, caption="Blue Channel", use_container_width=True)
51
52 # Grayscale + Colormap
53 st.subheader("Colormapped Grayscale Image")
54
55 colormap = st.selectbox(
56     "Choose a Matplotlib colormap",
57     ["viridis", "plasma", "inferno", "magma", "cividis", "hot", "cool", "gray"]
58 )
59
```

OUTLINE

- TIMELINE

Windows taskbar at the bottom shows the date and time as 02:03 PM, along with system icons for network, volume, and battery.

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Streamlit

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OUTLINE

TIMELINE

Welcome myimage.py

myimage.py > ...

58)

59

60 priyanka_gray = priyanka.convert("L")

61 priyanka_gray_np = np.array(priyanka_gray)

62

63 # Plot using matplotlib with colormap

64 fig, ax = plt.subplots(figsize=(6, 4))

65 im = ax.imshow(priyanka_gray_np, cmap=colormap)

66 plt.axis("off")

67

68 # DO NOT USE: plt.show()

69 # USE THIS INSTEAD:

70 st.pyplot(fig)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

streamlit

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C:\Users\DELL\Desktop\Streamlit>streamlit run myimage.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501

Network URL: http://192.168.1.12:8501

27°C Mostly cloudy

ENG 02:05 PM

Priyanka Sulaganti Image - Multi-Color Channel Visualizer





Original Priyanka Image

RGB Channel Visualization



Red Channel



Green Channel



Blue Channel

Colormapped Grayscale Image

Choose a Matplotlib colormap

gray

▼



